STAFF REVIEW · H446 022 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. An array stores indexed elements. Access an element by its index, for example values[2].

2. It stores a reference or index identifying another node, allowing nodes to be linked together.

3. LIFO: last in, first out (stack). FIFO: first in, first out (queue).

4. A: B, C; B: A; C: A, D; D: C. Each undirected edge is represented at both ends.

I Do — source worked steps

1. At index 0 — CODE:
30 < 40
current = left[0]
current = 1

2. At index 1 — CODE:
30 > 20
current = right[1]
current = 4

3. At index 4 — CODE:
data[4] = 30
found

4. Skipped nodes — CODE:
Did not inspect indexes 2 or 3.

We Do — guided CHECK

1. Complete one row
CODE:
index 0: Sligo
left = 1
right = 3

2. Complete the same row
CODE:
index 2: Cork
left = ?
right = ?
CODE:
index 2: Cork
left = null
right = null

3. Complete the path
CODE:
50 > 40 → ?
50 < 60 → ?
data = 50 → ?
CODE:
50 > 40 → right
50 < 60 → left
data = 50 → found

4. Complete the update
CODE:
left[1] = ?;
CODE:
left[1] = -1;

You Do — source mark schemes

1. Step 1: Assign an index to each node: 0=40, 1=20, 2=60, 3=10, 4=30.
CODE:
int[] data = {40,20,60,10,30};
int[] left = { 1, 3,-1,-1,-1};
int[] right = { 2, 4,-1,-1,-1};
Use -1 for a missing child. Root index = 0.

2. Target 10: 40 → left to 20 → left to 10. Found at index 3.
Target 60: 40 → right to 60. Found at index 2.
Target 99: 40 → right to 60 → right to -1. Not found.
CODE:
int current = 0;
int target = 10;
while (current != -1 && data[current] != target)
{
current = target < data[current]
? left[current] : right[current];
}
Console.WriteLine(current); // 3
Change target to 60 (output 2) and 99 (output -1).

3. Method: Record each town's array index first. In the Left and Right columns enter the index of its child, not the town name. Use null for an absent child when following the OCR table convention.
Known rows from the lesson’s guided examples: index 0, Sligo → Left 1, Right 3; index 2, Cork → Left null, Right null.
Check: Follow every non-null pointer to confirm it leads to the correct child. The complete eight-town tree is not reproduced in this lesson, so the remaining six rows must be checked against the original question diagram rather than guessed.

4. Step 1: Node 10 is at index 3. Its parent is node 20 at index 1.
Step 2: Since node 10 is a leaf, remove its parent's pointer to it.
CODE:
left[1] = -1;

// Search 10 again from root index 0:
// 40 → 20 → -1
Result: not found. The old value at data[3] is now unreachable from the root; a complete memory-management implementation would also mark the slot available for reuse.

Exit Ticket — source answer

1. True. The left and right fields identify the positions of child nodes rather than storing duplicate child values.